

# Product Carbon Footprint Study Report

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Product Name: integrated circuit production, logic type

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

## I. General Information

**Producer Name:** [Information provided by the user]

**Address:** [Information provided by the user]

**Legal Representative:** [Information provided by the user]

**Product Name:** integrated circuit production, logic type

**Product Function:** [Information provided by the user]

**Based on the Standard:** ecoinvent - EF v3.1

## II. Quantitative Purpose

1 kg integrated circuit production, memory type

## III. Scope of Quantification

### 1. Functional or Declaratory Units

Functional or declarative units in 1 kg integrated circuit production, logic type.

### 2. System Boundary

integrated circuit production, logic type, in China, in 2025

### 3. Cut-off Criteria

[Information provided by the user]

### 4. Time Scale

[Information provided by the user]

## IV. Inventory Analysis

### 1. Description of Data Sources

**Primary Data:** [Information provided by the user]

**Secondary Data:** ecoinvent

### 2. Principles and Procedures for Allocation

**Basis of Allocation:** [Information provided by the user]

**Allocation Procedure:** [Information provided by the user]

**Specific Allocations:** [Information provided by the user]

### 3. Data Quality Evaluation (Optional)

[Information provided by the user]

## V. Impact Assessment

### 1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

### 2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of integrated circuit production, logic type produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1526.311 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

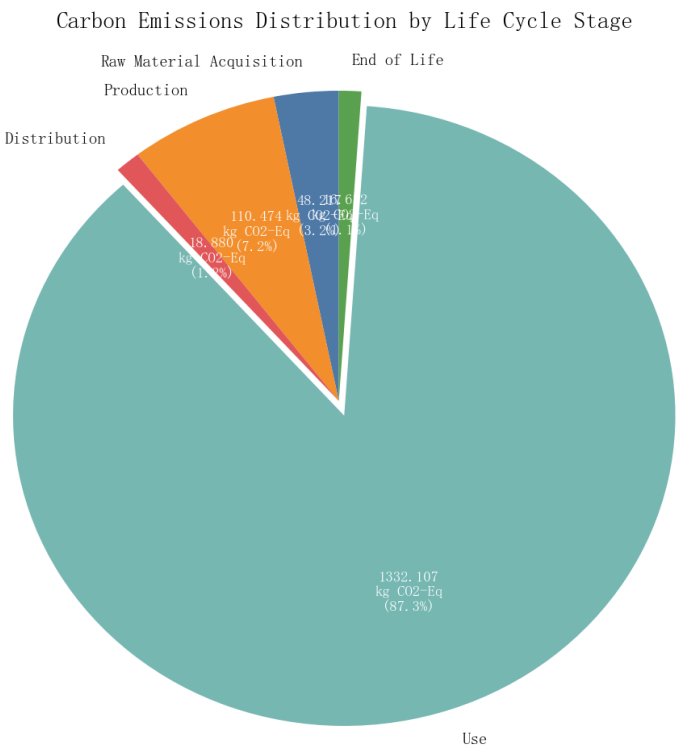


Table 1: Carbon Emissions by Life Cycle Stage

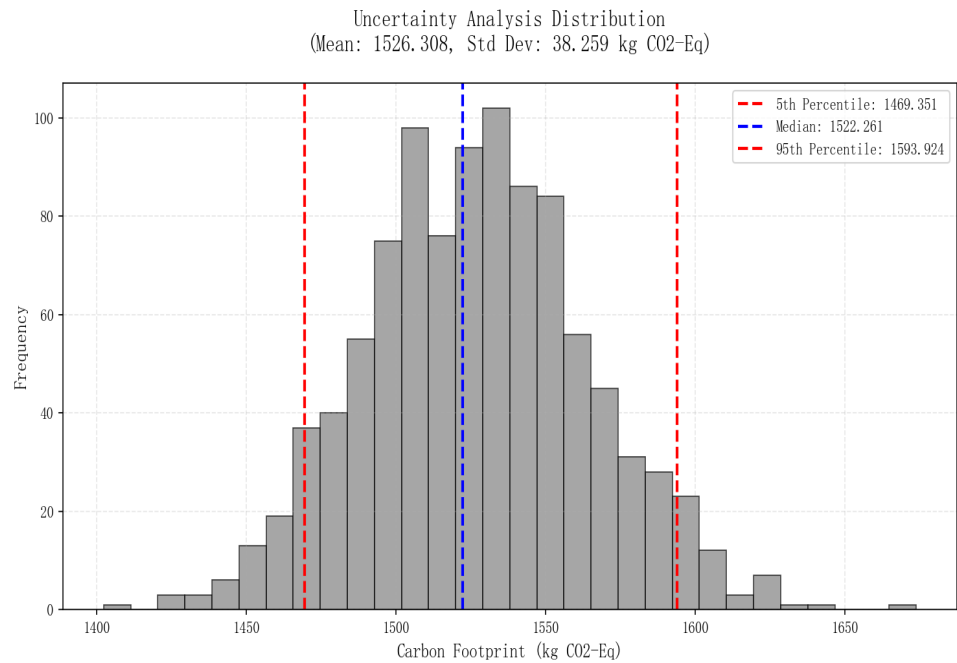
| Life Cycle Stage         | Carbon Footprint (kg CO2-Eq) | Percentage (%) |
|--------------------------|------------------------------|----------------|
| Raw Material Acquisition | 48.217                       | 3.16           |
| Production               | 110.474                      | 7.24           |
| Distribution             | 18.880                       | 1.24           |

|              |                 |               |
|--------------|-----------------|---------------|
| Use          | 1332.107        | 87.28         |
| End of Life  | 16.632          | 1.09          |
| <b>Total</b> | <b>1526.311</b> | <b>100.00</b> |

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



| Statistical Metric            | Value (kg CO2-Eq) |
|-------------------------------|-------------------|
| Mean                          | 1526.308          |
| Standard Deviation            | 38.259            |
| Coefficient of Variation (CV) | 2.51%             |
| Median                        | 1522.261          |
| 5th Percentile                | 1469.351          |
| 95th Percentile               | 1593.924          |

## VI. Interpretation of Results

### 1. Description of Results

The total carbon footprint of the integrated circuit (logic type) is 1526.311 kg CO<sub>2</sub>-Eq, representing a substantial environmental burden typical of high-performance semiconductor products. This value reflects a full cradle-to-grave assessment and highlights the product's intensive resource and energy demands across its lifecycle.

Lifecycle contributions reveal a pronounced dominance by the use phase, accounting for 1332.107 kg CO<sub>2</sub>-Eq or 87.3% of total emissions. Production contributes 7.2%, raw materials 3.2%, while distribution and end-of-life stages remain marginal at 1.2% and 1.1%, respectively. This distribution underscores that operational electricity consumption during the product's service life constitutes the primary emission driver, consistent with electronics characterized by prolonged high-power usage.

Major emission sources are therefore concentrated in the use stage, with secondary contributions from manufacturing energy inputs and material extraction. The low shares of distribution and end-of-life indicate limited leverage from logistics or disposal optimizations under current modeling.

### 2. Statement of Assumptions and Limitations

Key assumptions in this study include average global electricity grid mixes for the use phase, standard product lifetime of five years under typical computational workloads, and generic transport distances for distribution. End-of-life modeling presumes conventional recycling rates without region-specific waste management variations.

Limitations arise from reliance on secondary databases such as Ecoinvent, which may not fully capture proprietary fabrication processes or recent technological advancements in semiconductor nodes. Data quality is further constrained by aggregated industry averages rather than site-specific measurements, introducing uncertainty in emission factors for rare-earth materials and cleanroom energy use.

System boundary restrictions encompass cradle-to-grave coverage but exclude upstream infrastructure construction, indirect rebound effects from increased computing demand, and potential impacts beyond climate change such as water scarcity or toxicity. The assessment also omits use-phase scenarios under renewable-heavy grids or extreme high-performance computing applications.

### 3. Suggestions for Improvement

- Target the use phase by redesigning circuits for lower voltage operation and enhanced power gating to reduce electricity consumption by 20-30% during active workloads.
- Shift manufacturing facilities to renewable electricity sources and implement waste-heat recovery in cleanrooms to cut production-related emissions.
- Collaborate with suppliers to source lower-carbon raw materials, such as recycled silicon wafers, and optimize wafer yields to minimize material losses.

- Enhance data quality through primary data collection from fabrication plants and integration of real-time energy monitoring for more accurate use-phase modeling.
- Develop take-back programs with advanced disassembly to improve end-of-life recovery rates and explore carbon credits for verified emission reductions.